

Homework 04

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6.2

设混合咖啡中三种咖啡豆分别 x_1, x_2, x_3 千克, 则问题可表述为

$$\min 20x_1 + 28x_2 + 18x_3$$

$$\text{s.t. } x_1 \leq 500$$

$$x_2 \leq 600$$

$$x_3 \leq 400$$

$$x_1 + x_2 + x_3 = 1000$$

$$75x_1 + 85x_2 + 60x_3 \geq 75000$$

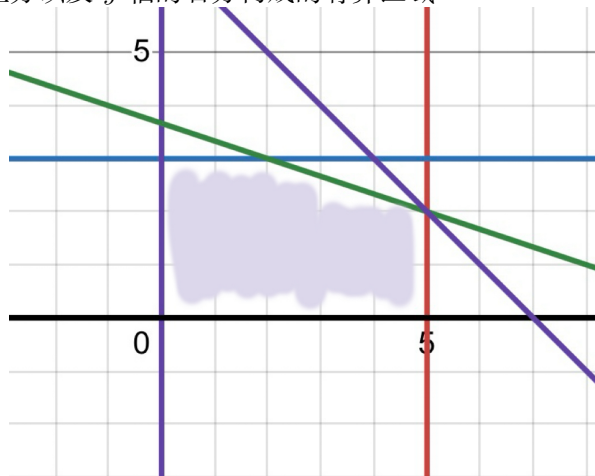
$$86x_1 + 88x_2 + 75x_3 \geq 80000$$

$$x_1, x_2, x_3 \geq 0$$

6.4

(1)

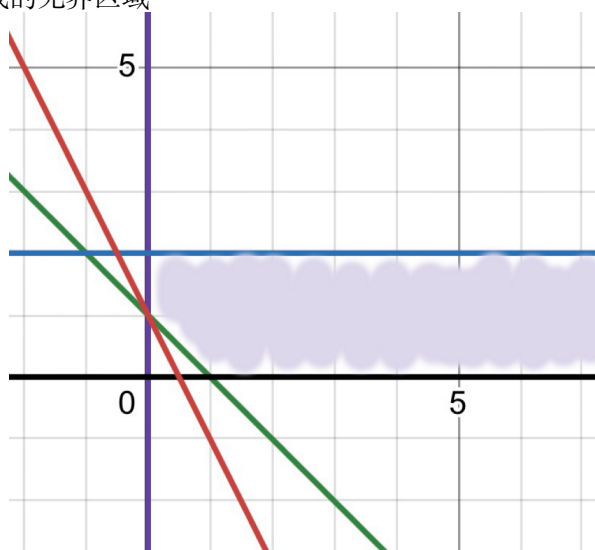
如图, 可行域是由直线 $x + 3y = 11$ 的左下方、直线 $x = 5$ 的左方、直线 $y = 3$ 的下方、 x 轴的上方以及 y 轴的右方构成的有界区域



最优点为 $x + y = 7$ 与该区域边界的交点, 最优解为 $x_1 = 5, x_2 = 2, z = x_1 + x_2 = 7$

(3)

如图，可行域是由直线 $x + y = 1$ 的右上方、直线 $y = 2$ 的下方、 x 轴的上方和 y 轴的右方构成的无界区域

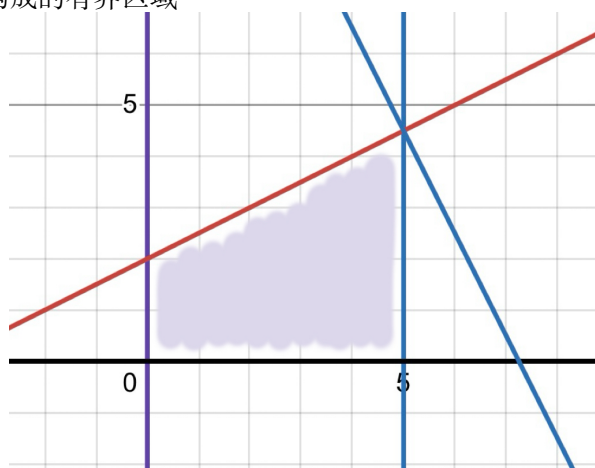


最优点为 $2x + y = 1$ 与该区域边界的交点，最优解为 $x_1 = 0, x_2 = 1, z = 2x_1 + x_2 = 1$

6.6

(1)

如图，可行域是由直线 $-x + 2y = 4$ 的右下方、直线 $x = 5$ 的左方、 x 轴的上方和 y 轴的右方构成的有界区域



最优点为 $2x + y = 14.5$ 与该区域边界的交点，最优解为 $x_1 = 5, x_2 = 4.5, 2x_1 + x_2 = 14.5$

(2)

标准形：

$$\min z = -2x_1 - x_2$$

$$\begin{aligned} \text{s.t.} \quad & -x_1 + 2x_2 + x_3 = 4 \\ & x_1 + x_4 = 5 \\ & x_j \geq 0, \quad j = 1, 2, 3, 4 \end{aligned}$$

$$A = \begin{bmatrix} -1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

所有基：

$$(P_1, P_2) = \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}, \quad x_1 = 5, x_2 = 4.5, x_3 = x_4 = 0, z = -2x_1 - x_2 = -14.5, \text{ 是可行基}$$

$$(P_1, P_3) = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}, \quad x_1 = 5, x_3 = 9, x_2 = x_4 = 0, z = -2x_1 - x_2 = -10, \text{ 是可行基}$$

$$(P_1, P_4) = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}, \quad x_1 = -4, x_4 = 9, x_2 = x_3 = 0, \text{ 不是可行基}$$

$$(P_2, P_3) = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}, \text{ 不是基}$$

$$(P_2, P_4) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad x_2 = 2, x_4 = 5, x_1 = x_3 = 0, z = -2x_1 - x_2 = -2, \text{ 是可行基}$$

$$(P_3, P_4) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad x_3 = 4, x_4 = 5, x_1 = x_2 = 0, z = -2x_1 - x_2 = 0, \text{ 是可行基}$$

其中最优解为 $\mathbf{X} = (5, 4.5, 0, 0)$

6.14

原始规划等价于

$$\begin{aligned} \max \quad & 3x_1 - 2x_2 + x_3 + 4x_4 - 4x_5 \\ \text{s.t.} \quad & x_1 + x_2 - x_3 - x_4 + x_5 \leq 6 \\ & -x_1 + 2x_2 - x_3 \leq -5 \\ & 2x_1 + x_2 - 3x_3 + x_4 - x_5 \leq -4 \\ & -2x_1 - x_2 + 3x_3 - x_4 + x_5 \leq 4 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

对偶规划为

$$\min \quad -6y_1 + 5y_2 - 4y_3 + 4y_4$$

$$\begin{aligned}\text{s.t. } & y_1 - y_2 + 2y_3 - 2y_4 \geq 3 \\ & y_1 + 2y_2 + y_3 - y_4 \geq -2 \\ & -y_1 - y_2 - 3y_3 + 3y_4 \geq 1 \\ & -y_1 + y_3 - y_4 \geq 4 \\ & y_1 - y_3 + y_4 \geq -4 \\ & y_1, y_2, y_3, y_4 \geq 0\end{aligned}$$

合并 y_3, y_4 及最后两个不等式, 得

$$\begin{aligned}\min & -6y_1 + 5y_2 - 4y_3 \\ \text{s.t. } & y_1 - y_2 + 2y_3 \geq 3 \\ & y_1 + 2y_2 + y_3 \geq -2 \\ & -y_1 - y_2 - 3y_3 \geq 1 \\ & -y_1 + y_3 = 4 \\ & y_1, y_2 \geq 0, \quad y_3 \text{ 任意}\end{aligned}$$